
EN 605P: Energy Systems Laboratory

Course Code	EN 605P
Course Name	Energy Systems Laboratory
L-T-P-C	0-0-3-2
Prerequisites	None
Intended for	Core for M.Tech. Energy Engineering
Distribution	Core
Semester	Odd/Even
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Preamble

The basic objective of this laboratory is to provide students a feel of actual systems and experience with some simple measurements in for characterising conventional as well as non-conventional energy sources and conversion methods.

Laboratory Modules

1. Study of properties of fuel oils & biomass, calorific value of Biomass samples
2. Testing of Gasifier: Effect of feedstock on heat output and pollutants
3. Energy conservation study of IC Engine
4. Piezo-electric energy harvesting: comparison of conversion efficiency of different materials and effect of vibration frequency on conversion efficiency
5. Electrical machines characteristics
6. Performance assessment of PV modules: I-V Characteristics and Efficiency of a Solar PV cell
7. Pollutant level measurement
8. Measurement of Solar Radiation and study of efficiency of a Flat Plate Solar Collector
9. Boiler, Pump & motor efficiency testing
10. Characterization of Refrigeration system
11. Study of fuel cells and measurement of their efficiency
12. Implementation of a solar powered pump
13. Design and implementation of a solar powered home with 24/7 availability of electricity
14. Visit of Hydro-electric power plant
15. Visit of Thermal power plant